

JARED ROBERT RICE

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Assistant Professor, Southwestern Adventist University

Education

PhD, Astronomy, 2018

University of Nevada, Las Vegas

Advisor: [Bing Zhang, PhD](#)

Dissertation: *Primordial black holes in the cosmological context and transient electromagnetic signatures from merging black hole binaries*

MS, Physics, 2012

Montana State University

Advisors: *Sachiko Tsuruta, PhD and Jiong Qiu, PhD*

BS, Astrophysics, 2008

University of California, Santa Cruz

Advisor: *Joel Primack, PhD*

Research interests

Electromagnetic counterparts of binary compact object mergers:

EM radiation mechanisms; relativistic blast waves; evolving synchrotron spectra; interaction of EM fields with gravitational waves; predicting observational EM and GW signatures of compact object mergers

X-ray and optical observations of high-energy astrophysical sources:

X-ray binaries in interacting and starforming galaxies; X-ray (*Chandra*) and optical (*HST*) data analysis

Gamma-ray bursts:

Multiwavelength GRB blast wave afterglows; GRB prompt emission; NS equation of state; gravitational wave emission

Very long baseline interferometry of supermassive black hole jet cores:

Probing black hole environments using the frequency-dependent synchrotron radio emission; astrometry of jet cores with the Very Long Baseline Array

Primordial black holes:

Accretion and evaporation histories of PBHs; PBHs as cosmic messengers; feasibility of detection

Publications

Metrics: Published papers: **19**; Citations: **300**; h-index: **9**; First author h-index: **3**; i10-index: **8**

[ADS entries](#), [ADS citation metrics](#)

1. (In prep, 2025) **Rice, J. R.**, Bamberger, A., and Zhang, B., *Transient electromagnetic signatures from merging supermassive black hole binaries*
2. (In prep, 2025) **Rice, J. R.**, Rangelov, B., Prestwich, A., Chandar, R., Bichon, L., and Boldt, C., *X-ray binaries in M51 II: individual sources*
3. (In prep, 2025) **Rice, J. R.**, Rangelov, B., Chandar, R., and Prestwich, A., *Optical counterparts to X-ray sources in nearby starburst galaxies*
4. (In prep, 2025) **Rice, J. R.**, Zavala, R.T., and Taylor, G.B., *Core shifts in compact symmetric objects*
5. Yang, X., Lü, H.-J., **Rice, J.**, Liang, E.-W., *Discovery of high-frequency quasi-periodic oscillation in short-duration gamma-ray bursts*, [Monthly Notices of the Royal Astronomical Society](#), **537**, 3 (2025)
6. Huang, J.-X., Lü, H.-J., **Rice, J.**, Liang, E.-W., *Spin, inclination, and magnetic field evolution of magnetar population in vacuum and plasma-filled magnetospheres*, [Physical Review D](#), **109**, 12 (2024)
7. Wang, S.-N., Lü, H.-J., Yuan, Y., Yuan, H.-Y., **Rice, J.**, Chen, M.-H., and Liang, E.-W., *Ejecta–Circumstellar Medium Interaction in High-density Environment Contribution to Kilonova Emission: Application to GRB 191019A*, [The Astrophysical Journal](#), **963**, 2 (2024)
8. Yuan, H.-Y., Lü, H.-J., **Rice, J.**, and Liang, E.-W., *Constraining the charge of a black hole with electromagnetic radiation from a black hole-neutron star system*, [Physical Review D](#), **108**, 8 (2023)

9. Yuan, H.-Y., Lü, H.-J., Li, Y., Zhang, B.-B., Sun, H., **Rice, J.**, Yang, J., and Liang, E.-W., *Probing the progenitor of high- z short-duration GRB 201221D and its possible bulk acceleration in prompt emission*, [Research in Astronomy and Astrophysics](#), **22**, 7 (2022)
10. Lü, H.-J., Yuan, H.-Y., Yi, T.-F., Wang, X.-G., Hu, Y.-D., Yuan, Y., **Rice, J.**, Wang, J.-G., Cao, J.-X., Kong, D.-F., Fernandez-García, E., Castro-Tirado, A. J., Lian, J.-S., Gan, W.-P., Wang, S.-Q., Xin, L.-P., Caballero-García, M. D., Fan, Y.-F., and Liang, E.-W., *GRB 211227A as a peculiar long gamma-ray burst from compact star merger*, [The Astrophysical Journal Letters](#), **931**, 2 (2022)
11. Huang, J.-X., Lü, H.-J., Liang, E.-W., and **Rice, J.**, *Gravitational-wave evolution of new-born magnetars with different deformed structure*, [Physical Review D](#), **105**, 10 (2022)
12. Yang, X., Lü, H.-J., Yuan, H.-Y., **Rice, J.**, Zhang, Z., Zhang, B.-B., and Liang, E.-W., *Evidence for gravitational lensing of GRB 200716C*, [The Astrophysical Journal Letters](#), **921**, L29 (2021)
13. **Rice, J. R.**, Rangelov, B., Prestwich, A., Chandar, R., Bichon, L., and Boldt, C., *X-ray binaries in M51 I: catalog and statistics*, [The Astrophysical Journal](#), **922**, 178 (2021)
14. **Rice, J. R.** and Zhang, B., *Growth of stellar mass black holes in dense molecular clouds and GW190521*, [The Astrophysical Journal](#), **908**, 59 (2021)
15. Lan, L., Lü, H.-J., Shen, J., **Rice, J.**, Li, L., and Liang, E.-W., *The properties of prompt emission in short GRBs with extended emission observed by Fermi/GBM*, [Monthly Notices of the Royal Astronomical Society](#), **492**, 3622 (2020)
16. Lan, L.; Lü, H.-J.; **Rice, J.**; and Liang, E.-W., *Constraining the nuclear equation of state via gravitational-wave radiation of short gamma-ray burst remnants*, [The Astrophysical Journal](#), **890**, 99 (2020)
17. Moravec, E. et al., *The early career perspective on the coming decade, astrophysics career paths, and the Decadal Survey process*, APC White Papers No. 8, [Bulletin of the American Astronomical Society](#), **51**, 8 (2019)
18. Lü, H.-J., Shen, J.; Lan, L., **Rice, J.**, Lei, W.-H., Liang, E.-W. *Diagnosing the remnants of binary neutron star merger from GW170817/GRB170817A event*, [Monthly Notices of the Royal Astronomical Society](#), **486**, 4479 (2019)
19. **Rice, J. R.**, *Primordial black holes in the cosmological context and transient electromagnetic signatures from merging black hole binaries*, [ProQuest Dissertations and Theses](#) (2018)
20. Lan, L., Lü, H.-J., Zhong, S.-Q., Zhang, H.-M., **Rice, J.**, Cheng, J.-G., Du, S.-S., Li, L., Lu, R.-J., and Liang, E.-W., *Characteristics of two-episode emission patterns in Fermi long gamma-ray bursts*, [The Astrophysical Journal](#), **862**, 155 (2018)
21. **Rice, J. R.** and Zhang, B., *Cosmological evolution of primordial black holes*, [Journal of High Energy Astrophysics](#), **13**, 22 (2017)
22. Lü, H.-J., Zhang, H.-M., Zhong, S.-Q., Hou, S.-J., Sun, H., **Rice, J.**, and Liang, E.-W., *Magnetar central engine and possible gravitational wave emission of nearby short GRB 160821B* [The Astrophysical Journal](#), **835**, 181 (2017)
23. Li, L.B., Zhang, Z.B., and **Rice, J.**, *Radio afterglow rebrightening: evidence for multiple active phases in gamma-ray burst central engines*, [Astrophysics and Space Science](#), **359**, 37 (2015)

Professional duties

- Panel Leveler for the first ever Dual Anonymous Peer Review (DAPR) *Chandra* Cycle 23 Proposal Review, June 21–24, 2021, [Chandra X-ray Observatory](#)
- Referee, 2021 – present, [Journal of Cosmology and Astroparticle Physics](#)
- Referee, 2019 – present, [Journal of High Energy Astrophysics](#)
- Contributed opinions to the [Early Career Astronomer and Astrophysicist Focus Session for the 2020 Decadal Survey](#), Washington, DC, October 8–9, 2018

Students mentored

- Ms. Angela Bamberger, *Southwestern Adventist University* (2023 – present)

Project: Transient electromagnetic signatures from merging black hole binaries

Textbook editing and illustrations

- Copyedited and produced numerous figures for Bing Zhang, *The Physics of Gamma-Ray Bursts*, Cambridge University Press (2018)
- Produced various diagrams for Thomas Banks, *Modern Quantum Field Theory: A Concise Introduction*, Cambridge University Press (2008)

Awards

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|---|--------|----|--------|
| – US Naval Observatory Flagstaff Station Colloquium Honorarium | (2018) | \$ | 250 |
| – Nevada NASA Space Grant Consortium Graduate Research Fellowship | (2016) | \$ | 21,000 |
| – UNLV Foundation Bigelow Travel Grant | (2014) | \$ | 4,000 |
| – MSU Outstanding Graduate Teaching Assistant Award | (2012) | | — |
| – NSF REU Research Grant | (2007) | \$ | 5,000 |
| – UCSC Crown College Undergraduate Research Fellowship | (2007) | \$ | 700 |

Skills

- General: [Python](#)/[iPython](#), [SAOImageDS9](#), [L^AT_EX](#), [IDL](#), [HTML](#)
- X-ray data analysis: [Chandra Interactive Analysis of Observations \(CIAO\)](#)
- Optical data analysis: [Image Reduction and Analysis Facility \(IRAF\)](#)
- Radio data reduction: [Astronomical Image Processing System \(AIPS\)](#)
- Gravitational wave data analysis: [GWpy](#) and [PyCBC](#)

Teaching

- Assistant Professor of Physics, Astronomy, and Mathematics (2022 – 2025), Southwestern Adventist University
- Guest lecturer, Introductory Astronomy (Fall 2020), Texas State University
- Guest lecturer, Graduate Astrophysics II (Spring 2017), University of Nevada, Las Vegas
- Guest lecturer, Introductory Astronomy (Fall 2015), University of Nevada, Las Vegas
- Graduate Teaching Assistant (2013 – 2018), University of Nevada, Las Vegas
- Adjunct Instructor of Physics (2012 – 2013), Miami University, Oxford, OH
- Adjunct Instructor of Physics (2013), Miami University Hamilton, Hamilton, OH
- Graduate Teaching Assistant (2009 – 2012), Montana State University
- Guest lecturer, Solar System Astronomy (Fall 2011), Montana State University

Conferences & Workshops

- [Yukawa Institute of Theoretical Physics International Molecule-type Workshop: Fast Radio Bursts: A Mystery Being Solved?](#), Online, February 8–19, 2021
- [ALMA Community Day Event](#), The University of Texas at Austin, Austin, TX, April 8, 2019
- [Early Career Astronomer and Astrophysicist Focus Session for the 2020 Decadal Survey](#), Washington, DC, October 8–9, 2018
- [16th Synthesis Imaging Workshop](#), New Mexico Tech, Socorro, NM, May 16–23, 2018
- [LIGO Open Data Workshop # 1](#), Caltech, Pasadena, CA, March 25–27, 2018
- [IAU 338: GW Astrophysics: Early Results from GW Searches and EM Counterparts](#), 2017
 - * Contributed talk: “*Radio afterglow of gravitation-driven plasma waves in SMBH binary mergers*”
- [Eighth Huntsville Gamma-Ray Burst Symposium](#), October 24–28, 2016
 - * Poster: “*Cosmological evolution of primordial black holes*”
- UNLV/Caltech Radio Transient Workshop, Las Vegas, April 11–12, 2016
- UNLV GRBs and Numerical Simulations Workshop, Las Vegas, September 9, 2015
- [European Week of Astronomy and Space Science](#), Geneva, Switzerland, 2014
- [IAU 307: New Windows on Massive Stars](#), Geneva, Switzerland, 2014
- UCSC Galaxy Formation and Evolution Workshop, UC Santa Cruz, August 6–10, 2007
- All-Wavelength Extended Groth Strip International Survey Meeting, UC Santa Cruz, December, 2006

Outreach

- SWAU “Totality at the Thomsen” (April 8, 2024) Eclipse Watch Party, Thomsen Observatory, Keene, TX
- Astronomy Day Volunteer (2010 – 2012), Museum of the Rockies, Bozeman, MT
- Instructor, Rocket Physics (2011), *MSU Peaks & Potentials* (elementary students)

Professional references

1. Bing Zhang, PhD, Distinguished Professor of Astrophysics; University of Nevada, Las Vegas
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2. Blagoy Rangelov, PhD, Assistant Professor of Astrophysics; Texas State University
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4. Darrell Pepper, PhD, Professor of Mechanical Engineering; University of Nevada, Las Vegas
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